

Ice

Ice covers one-tenth of Earth’s land surface, mostly in the polar regions. At earlier times in Earth’s history, when the climate was much cooler, ice covered an area up to three times larger than it does today.

Sea ice

Sea ice is frozen sea. It forms when the ocean’s surface freezes in winter. Where it lasts year round, it may be 20 ft (6 m) thick—elsewhere it is thinner. “Pancake ice” (right) is disks of sea ice up to 4 in (10 cm) thick.



Summer ice
The polar sea ice cover shrinks in summer, but some sea always remains under a layer of ice.

Winter ice As the weather gets colder, the polar sea ice spreads far beyond its summer limits.



Land ice

Thick ice gradually builds up on land as old, unmelted snow is compacted by layers of fresh snow and turned into ice. Antarctica’s ice sheet is up to 3 miles (4.8 km) thick.



Ice sheet A vast layer of land ice that has formed over thousands or even millions of years.

Ice shelf A floating extension of an ice sheet or glacier, usually hundreds of yards thick.